KSJ RESEARCH

Research Note Vol.1

Is AI Power Demand Real, and Does the Market Already Know?

Demand, ground truth, earnings and market pricing from official data — and what it means for IPO pricing

Shinjae Kang

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Demand is real, it shows up on the ground — earnings led the market (3-name net income, 28% of starting cap; multiple-led on 4-name operating income), but P/E also rose from 13.5x to 40.1x

IEA 2030E 945 TWh (2.3x 2024) · Virginia commercial sales +73% vs U.S. +10% · 4 names' market cap 7.7x combined (range 4.4x–26.1x, simple average 14.6x)

Key metrics [*1,*2,*3,*4,*5]

Key metric	Value	p.
2030E data-centre power	**945 TWh (2.3x)**	1
2035E outlook range	700~1,700 TWh	1
2025 growth: AI DC · DC · world	+50% · +17% · +3%	1
Virginia commercial index (2025)	**173 vs U.S. 110**	2
4 names' combined cap (2023-01→2026-10)	**7.7x**	4
Simple average / range	14.6x / 4.4x~26.1x	4
3-name earnings · multiple share	**61% · 39%**	4
3-name P/E (start-2023→latest)	**13.5 → 40.1x**	4
KTB 10y (start-2023→latest)	3.81 → 4.21%	5

Source: *1 IEA, Energy and AI (Apr 2025), executive summary *2 IEA, Key Questions on Energy and AI (Apr 2026) *3 U.S. Energy Information Administration (EIA), Electricity retail sales *4 Korea Exchange (KRX) Open API, KOSPI daily trading *5 FSS DART annual reports (consolidated)
Note: KSJ Research calculations; see footnotes on each page. Judgements are the author's own

2.3x

Data-centre power 2030E ÷ 2024 (IEA Base Case)

Demand is real — 945 TWh by 2030; AI data-centre demand +50%

- 2025: data centres +17%, AI data centres +50% vs world +3%
- Grew 12%/yr in 2017–2024; 1.5% of world power in 2024; U.S. 45%
- But the 2035 range is wide: 700–1,700 TWh (±42% vs base)

+73%

Virginia commercial power sales 2015→2025 (EIA)

It shows up on the ground — 63 index points ahead of the U.S. (+10%)

- CAGR Virginia 5.6% vs U.S. 0.9% (2015–2025)
- Last 3 years (2022→2025) CAGR 6.8% vs 2.4%
- U.S. commercial price +26% since 2015 (volume +9.8%)

7.7x

4 names, combined cap, 2023-01-02→2026-10-19 (range 4.4x–26.1x)

Earnings led — 61% of the gain for 3 names (ex. Doosan Enerbility) is earnings growth (3-name net-income basis, 28% of starting cap) · on 4-name operating income: earnings 31%, multiple 69% (multiple-led)

- Simple average 14.6x — Doosan Enerbility (4.4x), 72% of starting cap, drags the total
- 3-name aggregate P/E 13.5x → 40.1x (start-2023 ↔ FY2022A, latest ↔ FY2025A)
- Hyosung Heavy Industries has the largest earnings share (88%) — only LS ELECTRIC was multiple-led (57%)

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On 3 names net income, earnings led — only 28% of starting cap; on 4-name operating income: earnings 31%, multiple 69%

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An earnings-led sector, yet 40x P/E — hence Vol.2 data-centre PF, Vol.3 IPO pricing and overseas peers

App. **Appendix** A1–A4

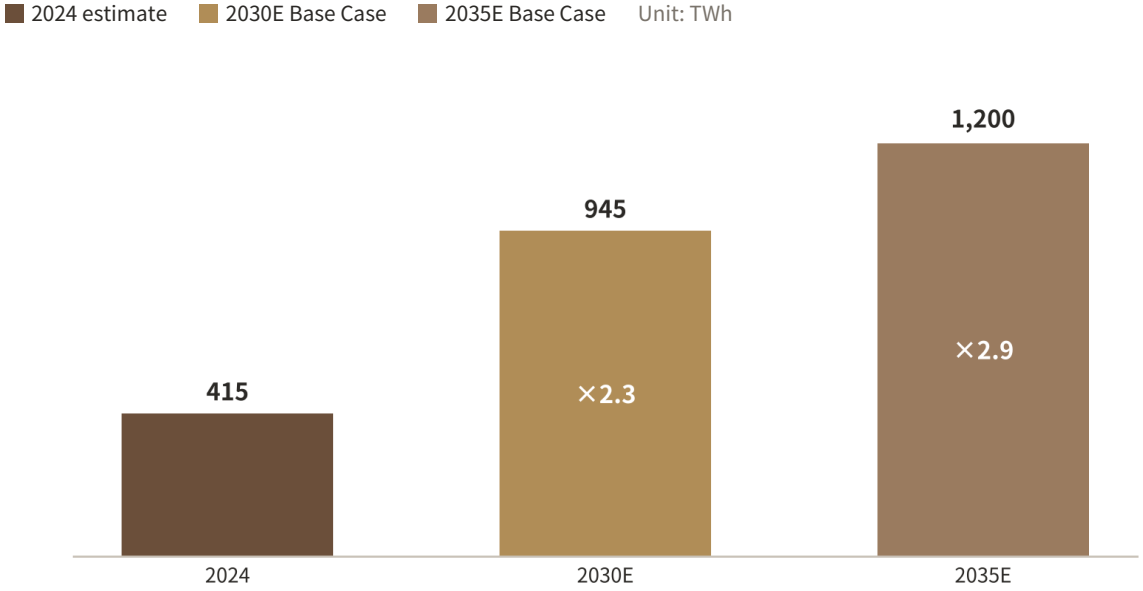
A1 Q&A · A2 Data notes (sources, datasets, dates, share-count check) · A3 Methodology · A4 Glossary

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Demand is real: data-centre power use reaches 2.3x by 2030; AI data-centre demand +50% in 2025

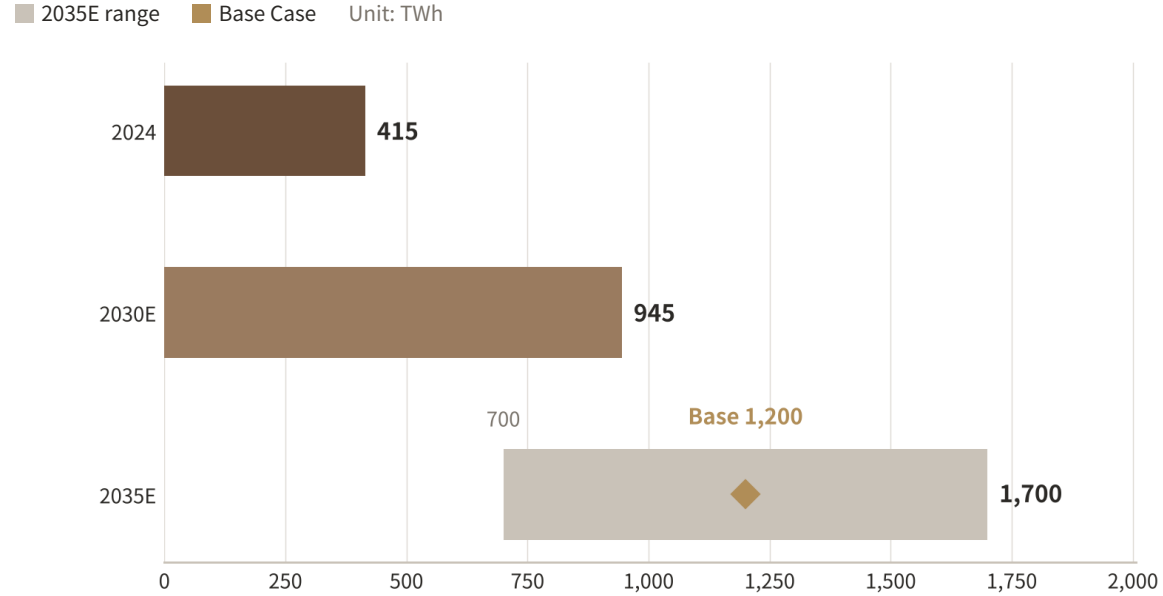
IEA Base Case: 415 TWh (2024) → 945 TWh (2030) → 1,200 TWh (2035) — but the 2035 range (700–1,700 TWh) is wide

Data-centre power use 2030E 945 TWh — 2.3x 2024 *1



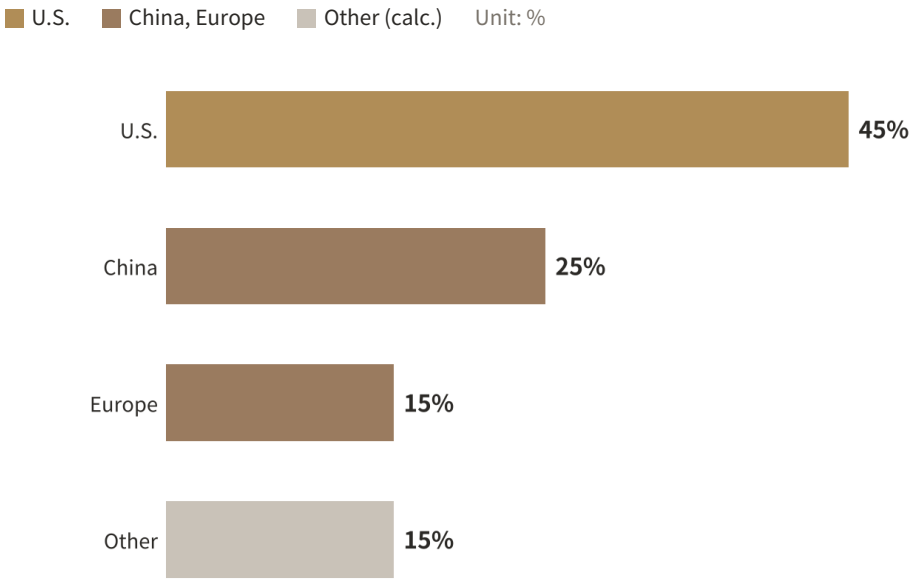
Source: *1 IEA, Energy and AI (Apr 2025), executive summary
Note: KSJ Research. Multiples inside bars = year ÷ 2024

2035 range 700–1,700 TWh — −42% to +42% vs Base Case *1



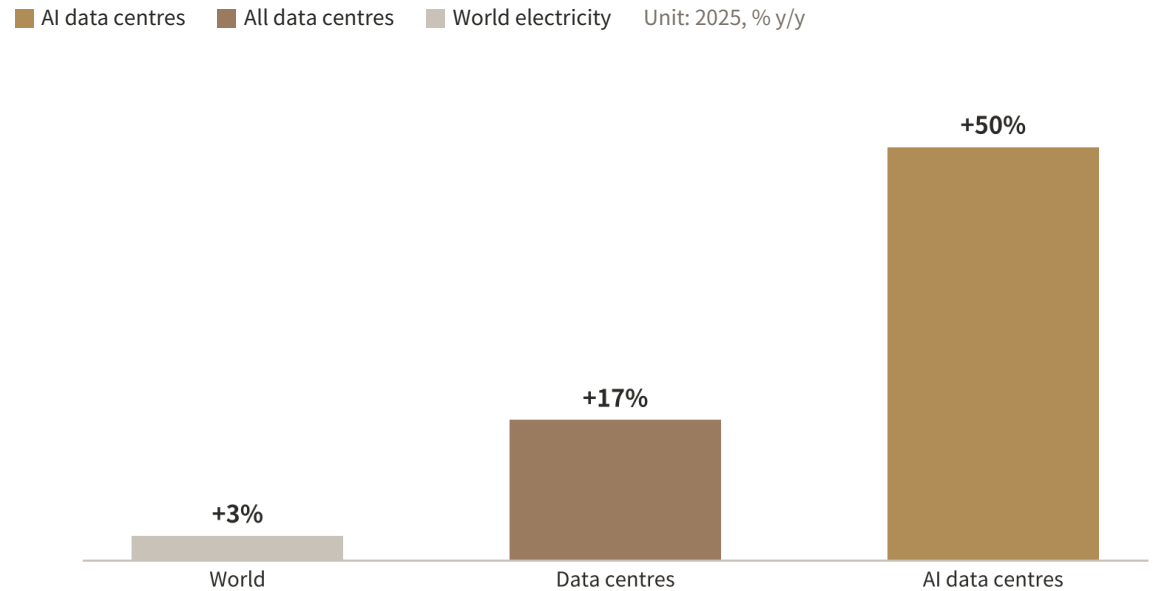
Source: *1 IEA, Energy and AI (Apr 2025), executive summary
Note: Range = IEA's 2035 low–high; % vs Base Case (KSJ Research)

The U.S. used 45% of data-centre power in 2024; U.S., China and Europe 85% *1



Source: *1 IEA, Energy and AI (Apr 2025), executive summary
Note: Other = 100 − U.S., China, Europe (KSJ Research). Data-centre power grew 12%/yr in 2017–2024; 1.5% of world power in 2024

AI data-centre demand +50% in 2025 — 17x the pace of world power (+3%) *1



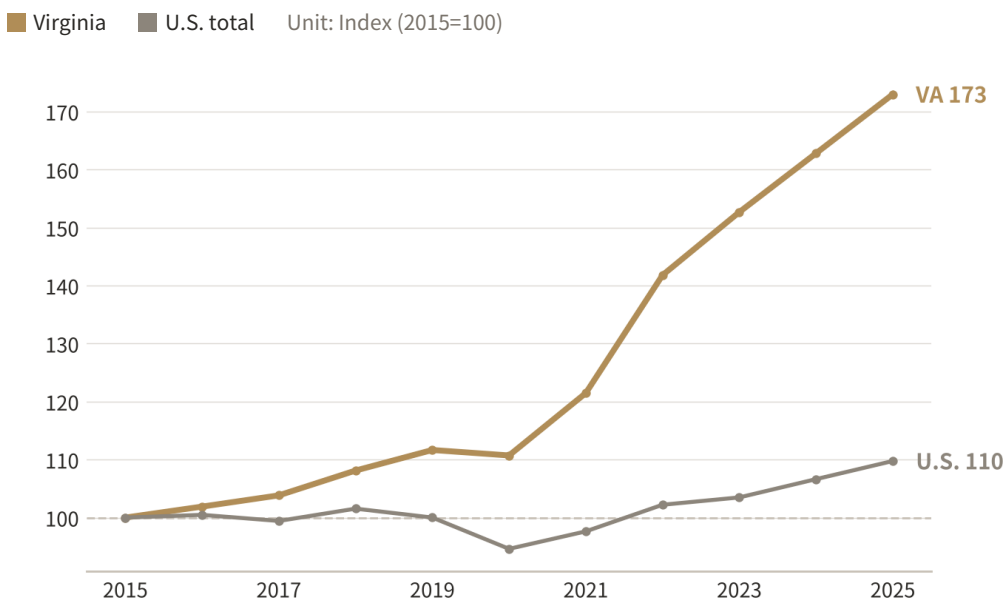
Source: *1 IEA, Key Questions on Energy and AI (Apr 2026)
Note: Multiple = 50 ÷ 3 (KSJ Research)

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It shows up on the ground: Virginia commercial sales +73% since 2015, 63 index points ahead of the U.S. (+10%)

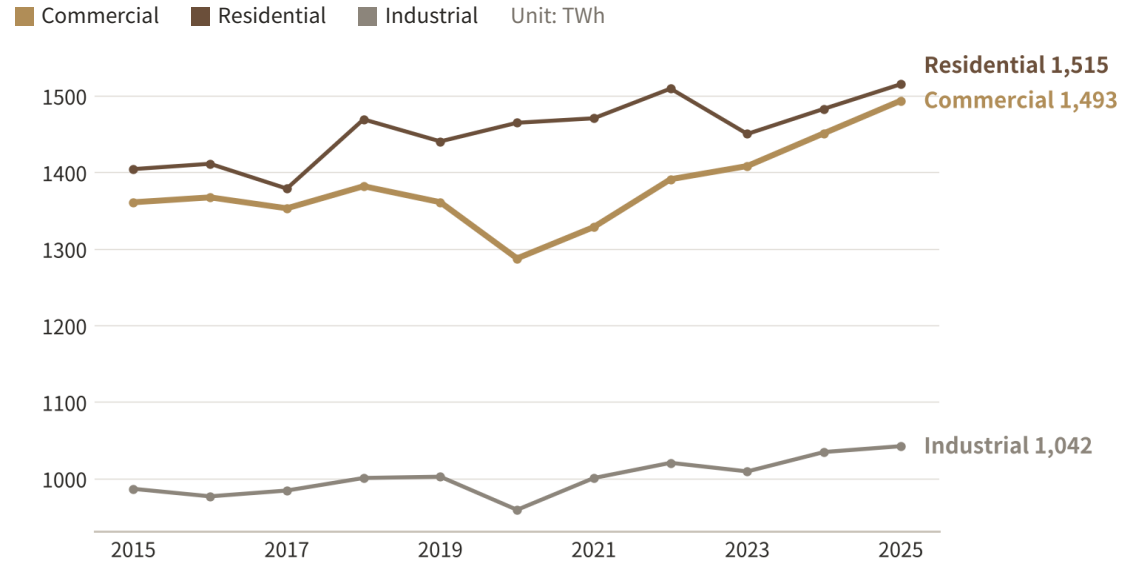
Virginia, known as a data-centre hub, grew commercial sales 4.7pp a year faster than the U.S. — U.S. commercial prices rose +26%

Virginia commercial sales +73% vs U.S. +10% — 63-index-point gap in 2025 *1



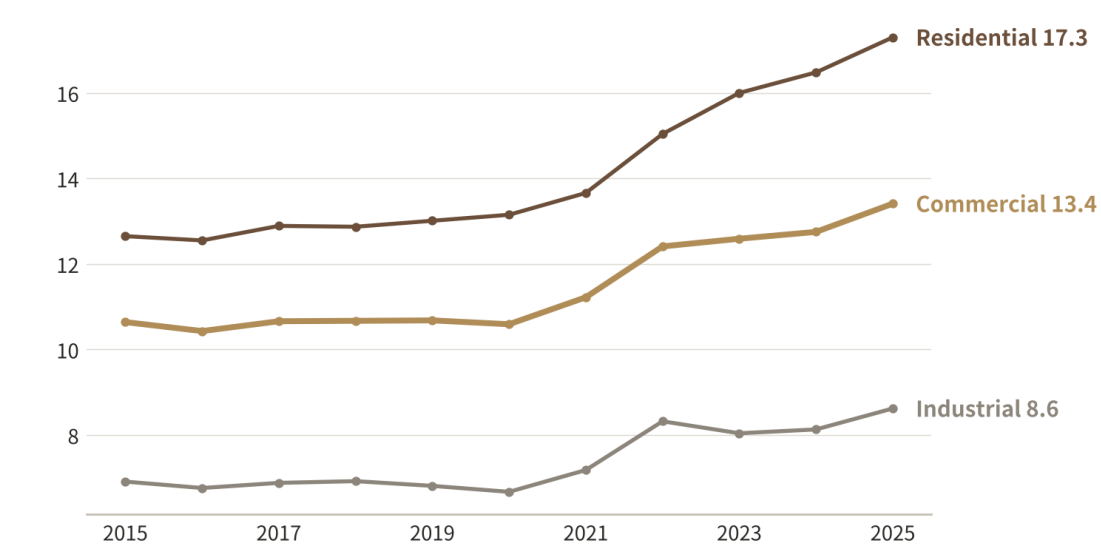
Source: *1 U.S. Energy Information Administration (EIA), Electricity retail sales
Note: Index = sales ÷ 2015 × 100 (KSJ Research). CAGR Virginia 5.6% vs U.S. 0.9%

U.S. sales by sector 2015→2025: Commercial +9.8% · Residential +7.9% · Industrial +5.6% *1



Source: *1 U.S. Energy Information Administration (EIA), Electricity retail sales
Note: Sales in million kWh ÷ 1,000 = TWh (KSJ Research). Includes the 2020 commercial dip

Commercial average price 10.64¢ → 13.41¢ (+26%) — faster rise since 2020 *1



Source: *1 U.S. Energy Information Administration (EIA), Electricity retail sales
Note: Commercial price CAGR 2015–2020 -0.1% → 2020–2025 4.8% (KSJ Research)

Last 3 years CAGR: Virginia 6.8% vs U.S. 2.4% *1

Unit: % y/y; gap in pp

Year	Virginia	U.S. total	Gap
2018	+4.1%	+2.1%	+2.0%p
2019	+3.3%	-1.5%	+4.8%p
2020	-0.8%	-5.4%	+4.6%p
2021	+9.7%	+3.2%	+6.5%p
2022	+16.7%	+4.7%	**+12.0%p**
2023	**+7.7%**	+1.2%	+6.4%p
2024	**+6.6%**	+3.0%	+3.6%p
2025	**+6.2%**	+2.9%	+3.3%p

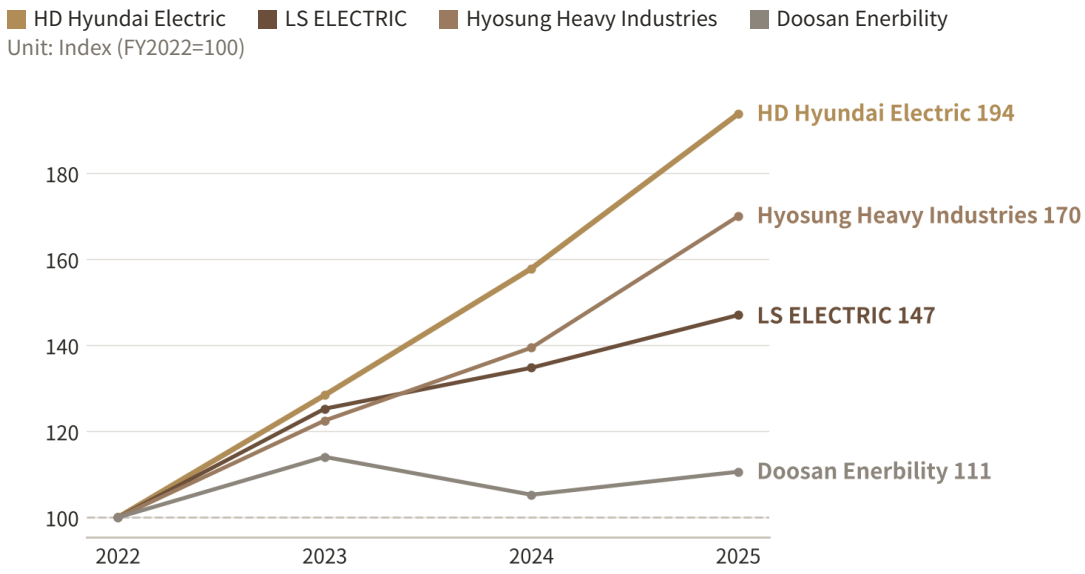
Source: *1 U.S. Energy Information Administration (EIA), Electricity retail sales
Note: Commercial sales growth y/y (KSJ Research). Shaded = last 3 years; gold = largest gap (2022)

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Earnings grew too: NI to parent of 3 names (Doosan Enerbility excluded: FY2022 net loss) 5.5x explains 61% of the market-cap gain (16.2x)

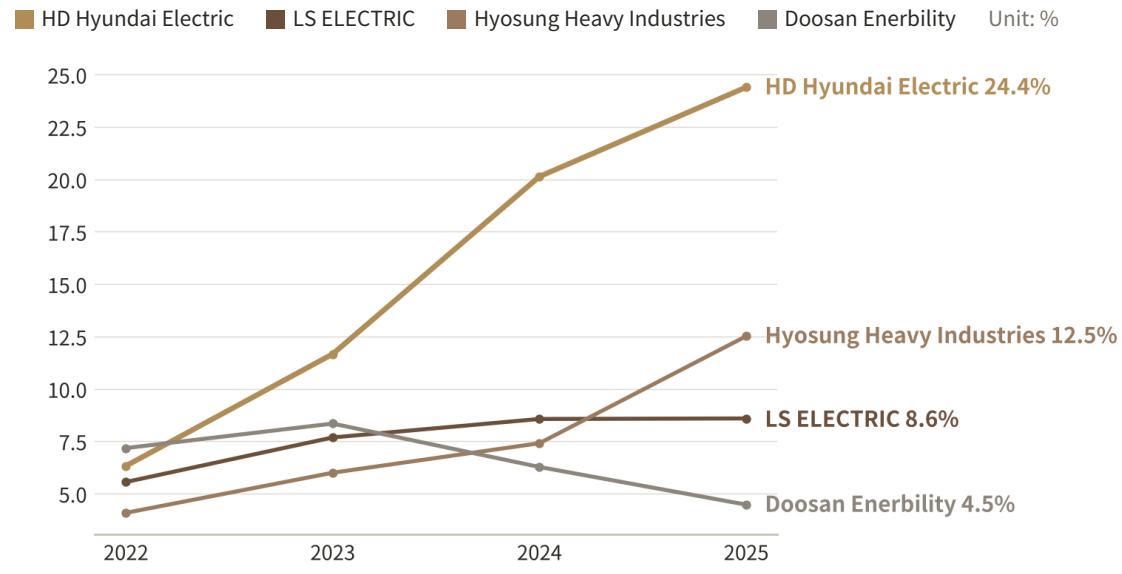
Combined revenue of 4 names +31%, operating profit of 4 names +87% (FY2022A→FY2025A, consolidated)

Revenue index (FY2022=100): HD Hyundai Electric highest at 194 · Doosan Enerbility 111 — combined +31% *1



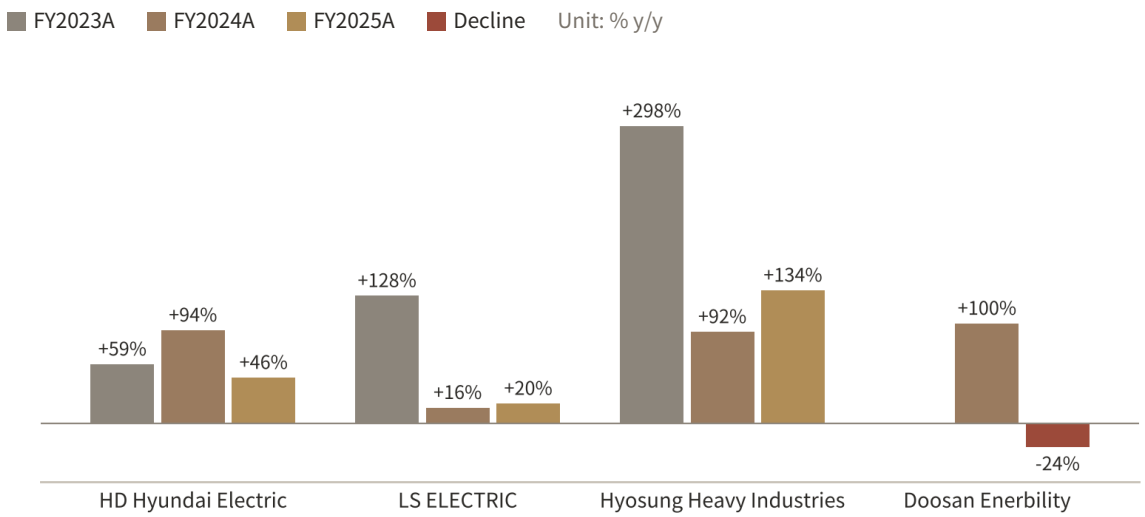
Source: *1 FSS DART annual reports (consolidated)
Note: Index = revenue ÷ FY2022A × 100 (KSJ Research). Consolidated; absolute values in the FY2025A table.

Operating margin FY2025A: HD Hyundai Electric highest at 24.4%; 3 of 4 improved *1



Source: *1 FSS DART annual reports (consolidated)
Note: Operating margin = operating profit ÷ revenue (KSJ Research).

NI to parent FY2022A→FY2025A: 3 names (Doosan Enerbility excluded: FY2022 net loss) combined 5.5x; Hyosung Heavy Industries leads at 17.8x; Doosan Enerbility turned profitable *1



Source: *1 FSS DART annual reports (consolidated)
Note: NI attributable to owners of the parent; no growth rate after a loss year (no bar). Combined multiple Doosan Enerbility excluded: FY2022 net loss. Hyosung Heavy Industries FY2022A: no NI-to-parent line in the filing, consolidated net income (incl. minorities) used.

FY2025A combined: revenue 320,717, OP 29,314, NI 16,239 (KRW 100mn) *1

Unit: KRW 100mn, %, x

Company	Revenue	OP	OPM	NI FY2022	NI FY2025	NI x
HD Hyundai Electric	40,795	9,953	24.4%	1,625	7,326	4.5x
LS ELECTRIC	49,658	4,264	8.6%	903	2,866	3.2x
Hyosung Heavy Industries	59,685	7,470	12.5%	291	5,199	**17.8x**
Doosan Enerbility	170,579	7,627	4.5%	-7,725	848	Turned profitable
Total	**320,717**	**29,314**	**9.1%**	**-4,906**	**16,239**	**5.5x (3 names)**

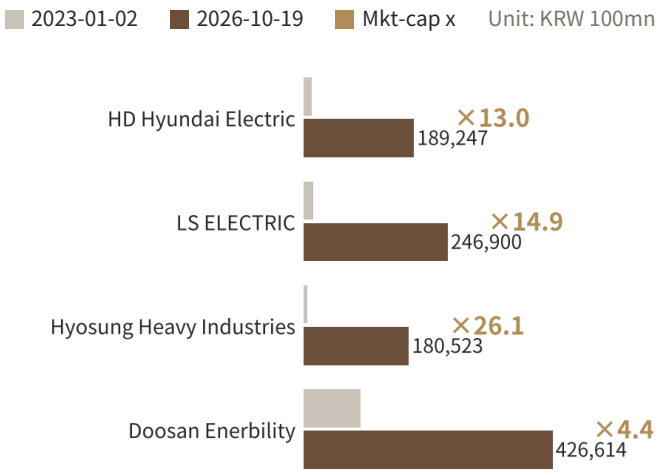
Source: *1 FSS DART annual reports (consolidated)
Note: Revenue, OP and OPM are FY2025A consolidated. NI x = FY2025A ÷ FY2022A; a label replaces the multiple when the base is ≤0. Combined multiple Doosan Enerbility excluded: FY2022 net loss (KSJ Research)

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On 3 names (ex. Doosan Enerbility) net income, earnings led (61%) — only 28% of starting cap; on 4-name operating income: earnings 31%, multiple 69% (multiple-led)

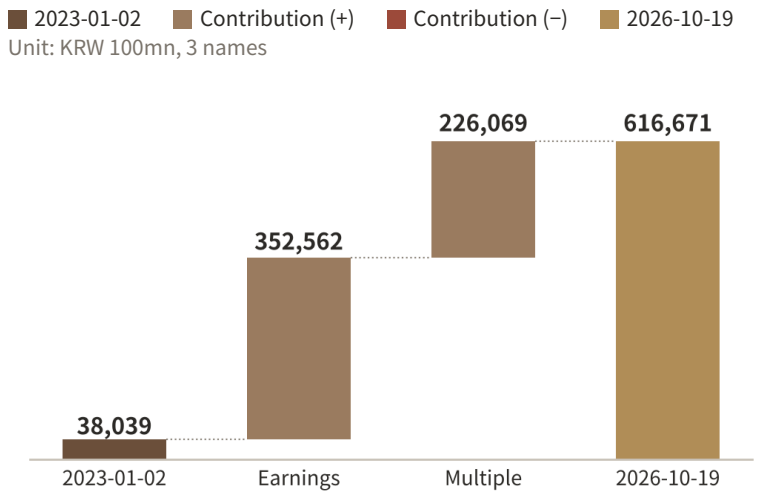
3-name aggregate P/E 13.5x → 40.1x · Doosan Enerbility (72% of starting cap) is excluded from the split: FY2022A net loss — separate from the 4-name combined 7.7x

Market cap: 7.7x combined · 14.6x simple average — 4.4x to 26.1x by name [*1,*2]



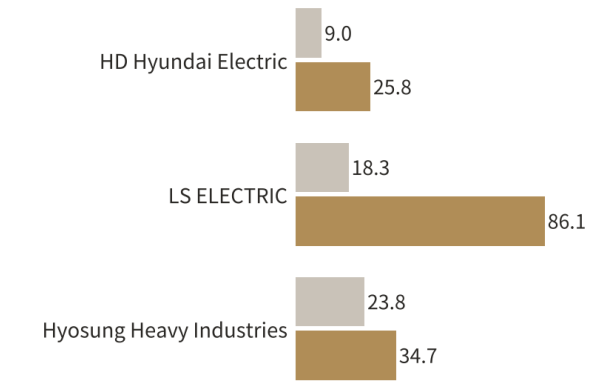
Source: *1 Korea Exchange (KRX) Open API, KOSPI daily trading *2 Company filing (stock split), author's check
Note: Market cap = close × listed shares at each date (KRX MKTCAP). The combined multiple is below the average because Doosan Enerbility (4.4x) was 72% of the starting cap. LS ELECTRIC: 5:1 split on 2026-04-13 — we use the market-cap multiple 14.9x, not the raw price multiple 3.0x (split-adjusted 14.9x).

Cap gain for 3 names (ex. Doosan Enerbility): 578,632 = earnings 352,562 + multiple 226,069 (KRW 100mn) [*1,*2]



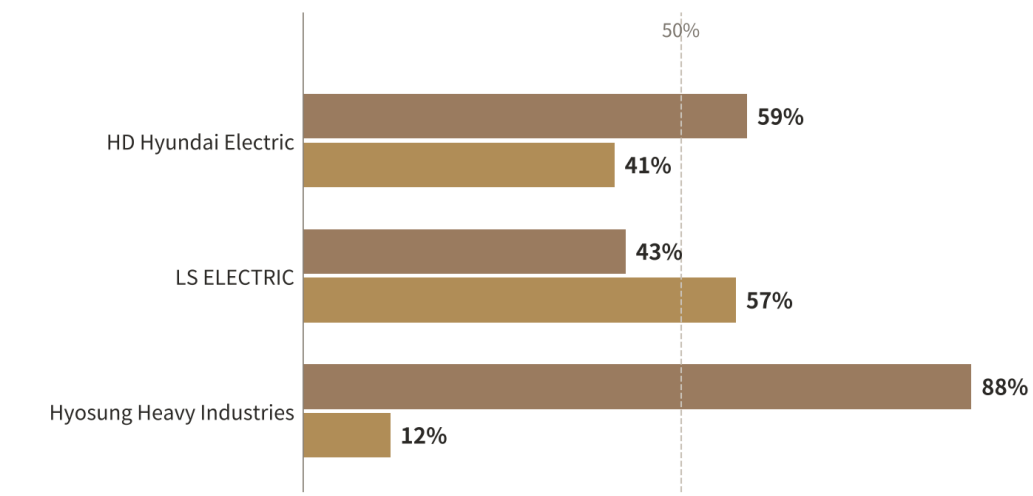
Source: *1 Korea Exchange (KRX) Open API, KOSPI daily trading *2 FSS DART annual reports (consolidated)
Note: Scope: 3 names (Doosan Enerbility excluded: FY2022A net loss). KRW split = Δcap × each term's log share. Log split Δln(cap) = Δln(NI) + Δln(P/E). Start-2023 cap ↔ FY2022A, latest cap ↔ FY2025A NI to parent. Only names with positive NI in both years are split (KSJ Research)

3-name P/E 13.5x → 40.1x; LS ELECTRIC highest at 86.1x [*1,*2]



Source: *1 Korea Exchange (KRX) Open API, KOSPI daily trading *2 FSS DART annual reports (consolidated)
Note: P/E = cap ÷ NI to parent. Doosan Enerbility: not shown, no starting P/E (FY2022A net loss). Including Doosan Enerbility, 4-name aggregate 64.2x (Doosan Enerbility 503x: denominator distortion from FY2025A NI of 848, KRW 100mn) Hyosung Heavy Industries FY2022A: no NI-to-parent line in the filing, consolidated net income (incl. minorities) used.

Hyosung Heavy Industries has the largest earnings share (88%) — only LS ELECTRIC was multiple-led (57%) [*1,*2]



Source: *1 Korea Exchange (KRX) Open API, KOSPI daily trading *2 FSS DART annual reports (consolidated)
Note: Earnings share = Δln(NI)/Δln(cap); multiple share = Δln(P/E)/Δln(cap); they sum to 100%. Dashed = 50%. Doosan Enerbility: not split (FY2022A net loss)

3-name aggregate P/E 13.5x → 40.1x: NI 5.5x × P/E 3.0x = cap 16.2x [*1,*2]

Unit: x; P/E start-2023 → latest; shares in %

Company	Cap x	NI x	P/E (start → latest)	Earnings	Multiple
HD Hyundai Electric	13.0x	4.5x	9.0 → 25.8	59%	41%
LS ELECTRIC	14.9x	3.2x	18.3 → 86.1	43%	**57%**
Hyosung Heavy Industries	26.1x	17.8x	23.8 → 34.7	88%	12%
Doosan Enerbility	4.4x	Turned profitable	—	Not split (FY2022A net loss)	
3-name total	**16.2x**	**5.5x**	**13.5 → 40.1**	**61%**	**39%**
Ref.: 4-name operating income	7.7x	1.9x (OP)	8.6 → 35.6 (cap/OP)	31%	**69%**

Source: *1 Korea Exchange (KRX) Open API, KOSPI daily trading *2 FSS DART annual reports (consolidated)
Note: P/E = cap ÷ NI (start-2023 ↔ FY2022A, latest ↔ FY2025A). Cap x = NI x × P/E x. Shaded = multiple-led. A label replaces the multiple when the base NI is ≤0. Ref. row = all four names have positive operating income, so the whole basket (100% of starting cap) is split the same way — the verdict depends on the earnings definition and scope

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An earnings-led sector, yet 40x P/E — hence Vol.2 data-centre PF (Korea 40 MW, U.S. 100 MW), Vol.3 IPO pricing and overseas peers

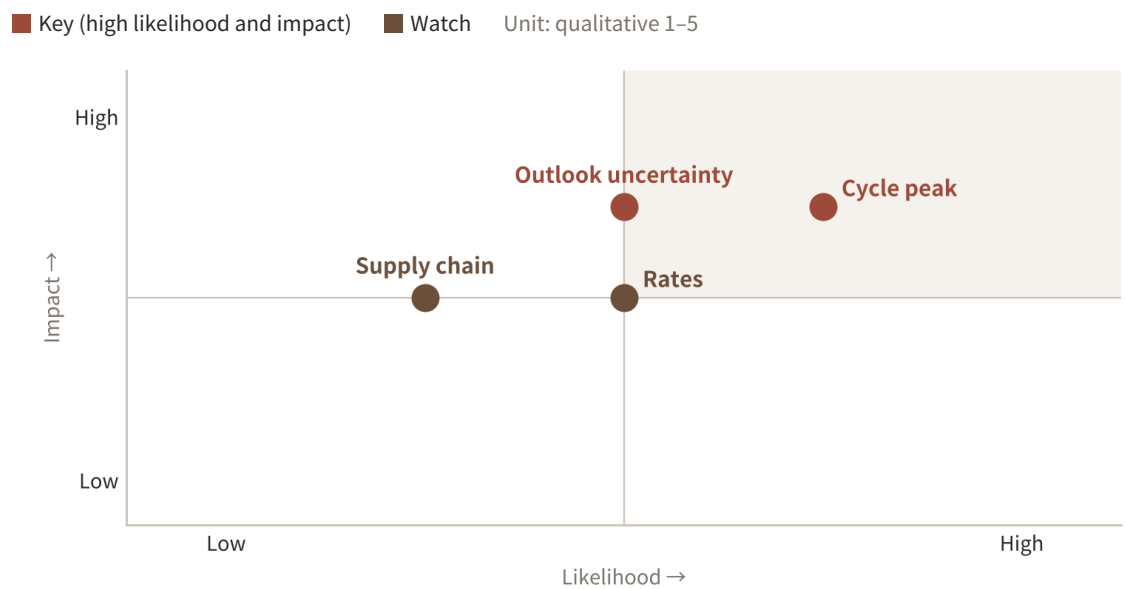
Aggregate P/E of 3 names (ex. Doosan Enerbility) 13.5x → 40.1x · three lenses (ECM pricing · data-centre PF · capacity) and four risks

Three lenses: data → judgement *1,*2,*3,*4

Lens	What the data says	Implication
ECM pricing	aggregate P/E of 3 names (ex. Doosan Enerbility) 13.5x → 40.1x; cap 7.7x combined (range 4.4x–26.1x, simple average 14.6x)	For 3 names (ex. Doosan Enerbility), earnings explain most of the gain, but P/E also rose 13.5x→40.1x. On 4-name operating income it is multiple-led, so no firm verdict. Rather than applying peer P/E as is, test earnings durability and apply an IPO discount
Data-centre PF	Virginia commercial sales 5.6%/yr; U.S. commercial price +26% (2015→2025)	With volumes rising (Virginia) and commercial prices rising (U.S. average, nominal), PPAs and grid-connection timing are the key PF cash-flow variables
Capacity options	Revenue +31% (FY2022A→FY2025A); top OPM 24.4%; IEA 2035 range 700–1,700 TWh	With a wide 2035 range, phased capacity additions that expand as demand is confirmed are preferable

Source: *1 Korea Exchange (KRX) Open API, KOSPI daily trading *2 FSS DART annual reports (consolidated) *3 U.S. Energy Information Administration (EIA), Electricity retail sales *4 IEA, Energy and AI (Apr 2025), executive summary
Note: Figures in the data column come from pages 1–4

Risk matrix: cycle peak and outlook uncertainty matter most *1,*2,*3



Source: *1 IEA, Energy and AI (Apr 2025), executive summary *2 Bank of Korea ECOS market rates (daily) *3 FRED, Federal Reserve Bank of St. Louis (DGS10)
Note: Anchors: IEA 2035 range ±42% vs base · aggregate P/E of 3 names (ex. Doosan Enerbility) 13.5x → 40.1x · KTB 10y 3.81→4.21% · UST 10y 3.79→5.09% (2023-01-02→2026-10-16) · supply chain is qualitative. Positions reflect the author's qualitative judgement

Next: Vol.2 data-centre PF (Korea 40 MW, U.S. 100 MW), Vol.3 IPO pricing and overseas peers

An earnings-led sector, yet 40x P/E: how should an IPO be priced?

- Vol.2 data-centre PF — hypothetical Korea 40 MW (KRW) and U.S. 100 MW (USD): debt capacity, DSCR and expansion option from a lender's view
- Vol.3 IPO pricing — hypothetical Target Co., assumptions disclosed with their rationale
- Vol.3 overseas peers — Korean peer P/E and EV/EBITDA vs overseas peers (SEC EDGAR)
- Fair value → IPO discount → indicative price range, Monte Carlo sensitivity

Note: Timing and method of Vol.2 and Vol.3 may change

Limitations of this note

- Ground data covers commercial sales only; data-centre load may be classified as industrial
- IEA figures are summary values entered manually and checked against the source
- The market window (Jan 2023–Oct 2026) and earnings window (FY2022A–FY2025A) do not match exactly
- P/E uses reported NI without one-off adjustments; thin earnings distort P/E
- The sample is four Korean names; overseas peer comparison comes in Vol.3
- The split and headline P/E cover the 3 names with positive base NI — excluding Doosan Enerbility (FY2022A net loss)

Note: To be addressed from Vol.2 onward

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Expected Q&A: answered with numbers

Questions likely in investment review or due diligence, with page references

11 expected questions [*1,*2,*3,*4,*5]

Question	Answer	Page
Isn't AI power demand overhyped?	IEA: 945 TWh in 2030E (2.3x 2024); AI data centres +50% in 2025 vs world +3%. The 2035 range (700–1,700 TWh) is wide	1
Isn't Virginia a special case?	It is one state. But commercial sales are +73% since 2015 (U.S. +10%), 6.8%/yr over 3 years (U.S. 2.4%). The cumulative gap has reached 63 points, but the annual growth gap narrowed from 12.0pp (2022) to 3.3pp (2025). Industrial classification is a stated limitation	2, 5
Do earnings support it?	NI of 3 names (Doosan Enerbility excluded: FY2022 net loss) rose 5.5x (FY2022A→FY2025A), explaining 61% of the market-cap gain (16.2x)	3, 4
Has the market priced it in?	For 3 names (ex. Doosan Enerbility), earnings explain 61% and multiples 39% (3-name net-income basis, 28% of starting cap); P/E rose 13.5x → 40.1x. But on 4-name operating income: earnings 31%, multiple 69% (multiple-led) — the verdict depends on the earnings definition and scope	4
Why is LS ELECTRIC's price up 3x but its market cap 15x?	A 5:1 split on 2026-04-13. The price move KRW 55,100 → 164,600 (2.99x) compares different share sizes; split-adjusted 2.99×5 = 14.9x, matching the cap multiple (shares 30,000,000→150,000,000)	4, A2
Which is right: 7.7x combined or 14.6x average?	Both — they answer different questions. Combined is a cap-weighted basket; the average weights names equally. Doosan Enerbility (4.4x) was 72% of the starting cap, pulling the combined figure down. Use combined for sector value, average for per-name re-rating	4
Why is Doosan Enerbility excluded from the split?	P/E = cap ÷ NI, and FY2022A NI was -7,725 (KRW 100mn), ≤0, so the starting P/E and log change are undefined (FY2025A: 848, turned profitable). At 72% of the starting cap it stays in the 4-name combined multiple (7.7x), but the split and headline P/E use 3 names. Its latest P/E of 503x reflects a thin denominator	3, 4
Why was LS ELECTRIC multiple-led?	Its cap rose 14.9x while NI rose 3.2x (rank 1 of 3, smallest first), and P/E went 18.3x → 86.1x, so the multiple share is 57%. Causes such as orders or business mix are outside this note's data	4
P/Es are high — is it a bubble?	Headline: aggregate P/E of 3 names (ex. Doosan Enerbility) 13.5x → 40.1x. P/E rose even as earnings grew; whether it is a bubble depends on earnings durability	4, 5
What comes next?	Vol.2 data-centre PF (Korea 40 MW, U.S. 100 MW), Vol.3 IPO pricing and overseas peers. Vol.2 sizes AI data-centre PF debt from a lender's view; Vol.3 sets an IPO price range reflecting this note's P/E levels and compares overseas peers	5
How can this be reproduced?	Full code and data sources are public on GitHub: https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio	A3

Source: *1 IEA, Energy and AI (Apr 2025), executive summary *2 U.S. Energy Information Administration (EIA), Electricity retail sales *3 Korea Exchange (KRX) Open API, KOSPI daily trading *4 FSS DART annual reports (consolidated) *5 Company filing (stock split), author's check
Note: Numbers match the body; shaded = data-treatment questions

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Data notes: sources, datasets, dates and the market-cap check

LS ELECTRIC split treatment stated · market cap uses listed shares at each date · overseas peers in Vol.3

Data sources · datasets · as-of dates

Source	Dataset	Items	As of	Status
DART · FSS	Single-company full financial statements (annual report, consolidated first)	Revenue, OP, NI to parent (4 peers)	2026-10-20	Collected
KRX · Korea Exchange	KOSPI daily trading data	Close, market cap, listed shares	2026-10-20	Cross-check open
EIA · U.S. EIA	Electricity retail sales (by state and sector, annual)	Retail sales and price by state/sector	2026-10-20	Collected
ECOS · Bank of Korea	Market interest rates (daily), KRW exchange rates	KTB 3y/10y	2026-10-20	Collected
FRED · St. Louis Fed	10-Year Treasury Constant Maturity (DGS10), KRW/USD (DEXKOUS)	DGS10, DEXKOUS	2026-10-20	Collected
IEA reports	Energy and AI (Apr 2025), Key Questions on Energy and AI (Apr 2026)	Data-centre power outlook and growth	2026-10-21	Verified vs source

Note: As-of = data collection date (IEA: date checked against the source). Overseas peer comparison comes in Vol.3. KRX closes differ from the secondary source (FinanceDataReader) by −4.6% to −8.3%, unresolved (likely a date mismatch; re-check pending)

Market-cap check: listed shares at each date (2023-01-02 · 2026-10-19) *1,*2

Company	Shares 2023-01-02	Shares 2026-10-19	Corporate action	Price x (raw)	Split-adj. price x	Cap x	Cap check
HD Hyundai Electric	36,047,135	36,047,135	No change	12.98x	12.98x	12.98x	0.00%
LS ELECTRIC	30,000,000	150,000,000	5:1 split 2026-04-13 (ratio matches)	2.99x	14.94x	**14.94x**	0.00%
Hyosung Heavy Industries	9,324,548	9,324,548	No change	26.06x	26.06x	26.06x	0.00%
Doosan Enerbility	638,308,033	640,561,146	Shares +0.35% — to be checked in filings (negligible effect)	4.37x	4.37x	4.38x	0.00%

Source: *1 Korea Exchange (KRX) Open API, KOSPI daily trading *2 Company filing (stock split), author's check
Note: Cap = KRX market cap (close × listed shares at each date). Check = close × shares ÷ market cap − 1. Split-adjusted price x = price x × split ratio. Price comparisons use split-adjusted prices. Corporate actions checked against company filings

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Methodology: official data, validated and calculated into this report

Every figure is calculated from official-source data, with sources and dates in the footnotes

Official sources → validation → calculation → report [*1]

1. Official sources

- DART
- KRX
- EIA
- ECOS
- FRED
- IEA reports (verified)

2. Validation

- Cap = close × listed shares
- Stock-split check
- Identifier check
- Units and FX

3. Calculation

- Indices, CAGR
- Log decomposition
- $\Delta\ln(\text{cap}) = \Delta\ln(\text{NI}) + \Delta\ln(\text{P/E})$
- Exclude base NI ≤ 0

4. Report

- Conclusion titles
- Korean & English
- Sources and dates

Built with an AI coding tool (Claude Code); question design, scope decisions and number verification by the author.

Source: *1 KSJ Research
Note: Full code and data sources: GitHub https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio

Glossary: units and calculations used in this note

Units: TWh, KRW 100mn, %, x

Glossary

Term	Definition
TWh	Terawatt-hour, one billion kWh; annual electricity use
Base Case	IEA's central scenario; the range is IEA's low–high
Commercial sector	EIA retail-sales class; data centres may be commercial or industrial by state
Index (2015=100)	Year value ÷ 2015 value × 100
CAGR	Compound annual growth = (end ÷ start)^(1/years) − 1
Market cap	Close × listed shares at that date (KRX MKTCAP)
Combined multiple	Sum of caps (end) ÷ sum (start); cap-weighted
Simple-average multiple	Average of each name's multiple; equal-weighted
Stock split	One share becomes several; cap unchanged, price adjusts → compare split-adjusted prices
NI to parent	Net income attributable to owners of the parent (DART consolidated)
PER	Market cap ÷ NI to parent
Log decomposition	Δln(cap) = Δln(NI) + Δln(P/E), an identity since cap = NI × P/E
Earnings / multiple share	Each term ÷ Δln(cap)
ECM	Equity capital markets: IPOs, follow-ons
Indicative price range	Price range set before book-building
PF	Project finance, repaid from project cash flows

Note: Shaded = core calculation terms in this note

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https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio

Contact: toonword@naver.com